

Do these tasks need both views?

Single view is enough · extra reasoning does not change it

Qwen3.5-9B — does reasoning help fuse the two views?

Overall · 9 urbanization tasks · Qwen3.5-9B

strategy	full	sat	sv	blind	full–best1	full–blind
current (direct)	48.6	44.4	50.9	40.9	-2.3	+7.7
think16k	49.8	46.1	51.8	41.0	-2.0	+8.8
multistep (4-turn)	51.7	49.4	53.9	43.4	-2.2	+8.3

full–best1 = full – max(sat, sv). Negative $\Rightarrow$ the second view does *not* help.

Street-view alone (sv) $\geq$ both views (full) for every strategy — thinking and multi-step raise the ceiling but lift the single view in lockstep, so fusion never starts working.

Per task · current (direct) · Qwen3.5-9B

task	full	sat	sv	blind	full–best1
Land use	53.0	50.1	52.7	43.5	+0.3
Building height	61.1	47.9	64.7	45.8	-3.6
Urban density	38.5	39.2	44.9	40.4	-6.4
Junction type	62.6	48.8	62.6	48.5	+0.0
Green space	36.5	35.5	37.9	64.5	-1.4
Amenity richness	36.8	36.1	33.5	19.5	+0.7
Road type	47.5	40.1	57.5	34.9	-10.0
Road surface	82.2	78.9	87.1	67.7	-4.9
Transit density	32.1	30.9	30.4	27.3	+1.2
OVERALL	48.6	44.4	50.9	40.9	-2.3

task	full	sat	sv	blind	full–best1
Land use	51.8	45.1	51.5	51.5	+0.3
Building height	62.6	51.6	61.6	38.4	+1.0
Urban density	38.7	38.0	43.9	32.3	-5.2
Junction type	57.5	54.8	59.0	32.3	-1.5
Green space	41.4	38.4	49.8	59.1	-8.4
Amenity richness	36.3	37.3	34.9	20.9	-1.0
Road type	58.4	50.4	64.4	34.4	-6.0
Road surface	83.2	75.2	84.8	94.4	-1.6
Transit density	31.6	33.3	31.6	26.6	-1.7
OVERALL	49.8	46.1	51.8	41.0	-2.0

Per task · multistep (4-turn) · Qwen3.5-9B

task	full	sat	sv	blind	full–best1
Land use	49.4	50.1	53.0	50.6	-3.6
Building height	61.1	54.7	58.4	43.2	+2.7
Urban density	41.6	43.7	45.4	35.9	-3.8
Junction type	63.5	59.3	62.9	46.4	+0.6
Green space	64.5	62.1	65.0	68.5	-0.5
Amenity richness	37.1	38.7	36.1	25.9	-1.6
Road type	53.4	48.5	63.2	33.3	-9.8
Road surface	89.8	80.9	89.4	83.8	+0.4
Transit density	29.7	26.8	31.4	27.8	-1.7
OVERALL	51.7	49.4	53.9	43.4	-2.2

The full models on the benchmark

Overall · full mode · 4 models

model	full	blind	vision gain
Gemma-4-12B	50.3	23.5	+26.8
Qwen2.5-VL-7B	47.2	38.3	+8.9
Qwen3.5-4B	49.0	38.9	+10.1
Qwen3.5-9B	51.1	38.1	+13.0

vision gain = full – blind. Vision genuinely helps — but (Story 1) it is *one-view* vision.

Per task · full-mode accuracy · 4 models

task	Gemma-4-12B	Qwen2.5-VL-7B	Qwen3.5-4B	Qwen3.5-9B
Land use	47.3	55.6	55.6	53.2
Building height	61.6	57.9	61.6	56.8
Urban density	37.3	41.8	36.3	39.0
Junction type	66.5	57.5	50.0	55.7
Green space	58.6	81.8	41.4	36.9
Amenity richness	35.6	31.8	35.4	36.3
Road type	65.3	50.6	57.0	53.0
Road surface	78.2	90.8	91.4	85.5
Transit density	28.0	28.0	34.2	31.4
OVERALL	50.3	47.2	49.0	51.1

OVERALL is over the whole benchmark (all topics); per-task rows are the 9 urbanization tasks.

Takeaway

- For the 9 urbanization tasks, a **single view is enough**: street-view alone matches or beats the two-view setup ($\text{full} - \text{best1} \leq 0$) for every strategy and every task.
- **Adding reasoning does not fix it**: 16k-token thinking and a 4-turn observe→reason→ self-check→commit conversation raise accuracy, but the single view rises with it.
- Across all 4 models vision *does* help ($\text{full} > \text{blind}$) — but it is one-view vision.